MAE 3240

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Problem set number: 3

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I consulted the class slides.
I went to office hours.

Problem 1

Known:
- radius R
- uniform volumetric heat generation rate $\dot{q}$
- T_∞, h
- $T(x,t)$

Properties:
- h
- K
- ρ
- c_p

Assumptions:
- thermal radiation neglected
- radius is very small
- temperature along radial direction is uniform

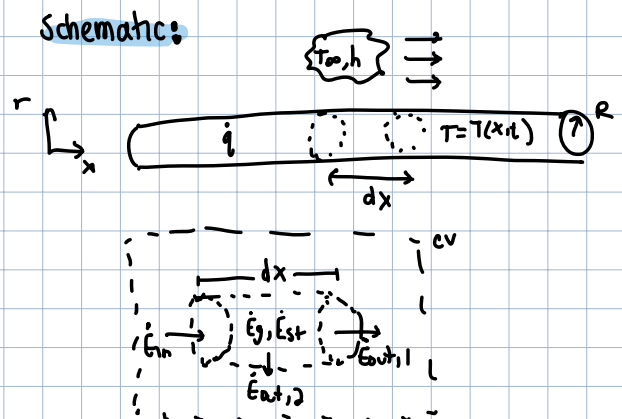
Find:

(a) consider a differential cylinder with a height of dx as the control volume. write the energy balance equation

(b) derive the governing differential equation that describes the temperature profile of the rod and simplify

(c) write the proper boundary & initial conditions for the governing dif. eq.

Schematic:



Analysis:

(a) $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = \dot{E}_{st}$

$$\boxed{\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}}$$

conduction $\dot{E}_{in} = -KA\frac{\partial T}{\partial x} = -K(\pi R^2)\frac{\partial T}{\partial x}\Big|_x \Rightarrow \dot{E}_{in} = -K(\pi R^2)\frac{\partial T}{\partial x}\Big|_x$

conduction $\dot{E}_{out,1} = -KA\frac{\partial T}{\partial x} = -K(\pi R^2)\frac{\partial T}{\partial x}\Big|_{x+dx} \Rightarrow \dot{E}_{out,1} = -K(\pi R^2)\frac{\partial T}{\partial x}\Big|_{x+dx}$

convection $\dot{E}_{out,2} = hA(T-T_\infty) = h(2\pi R\,dx)(T-T_\infty) \Rightarrow \dot{E}_{out,2} = h(2\pi R\,dx)(T-T_\infty)$

$\dot{E}_g = q'''V = \dot{q}A\,dx = \dot{q}(\pi R^2)\,dx \Rightarrow \dot{E}_g = \dot{q}(\pi R^2)\,dx$

$$\dot{E}_{st} = \rho V c \frac{\partial T}{\partial t} = \rho\,(\pi R^2 dx)\, c_p \frac{\partial T}{\partial t} \Rightarrow \dot{E}_{st} = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t}$$

$$\Rightarrow \underbrace{-k\pi R^2 \left.\frac{\partial T}{\partial x}\right|_x + k\pi R^2 \left.\frac{\partial T}{\partial x}\right|_{x+dx}} + \dot{q}\,(\pi R^2)\,dx - h\,(2\pi R\,dx)(T - T_\infty) = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t}$$

$$\Rightarrow k\pi R^2 \underbrace{\left[\left.\frac{\partial T}{\partial x}\right|_{x+dx} - \left.\frac{\partial T}{\partial x}\right|_x \right]}$$

$f(x+dx) - f(x)$ where $f(x) = \frac{\partial T}{\partial x}$

$\Rightarrow f(x+dx) - f(x) \approx \frac{\partial f}{\partial x} dx$

$\Rightarrow \frac{\partial}{\partial x}\left(\frac{\partial T}{\partial x}\right) dx$

$$\Rightarrow \boxed{\frac{\partial}{\partial x}\left(k\pi R^2 \frac{\partial T}{\partial x}\right) dx + \dot{q}(\pi R^2)\,dx - h\,(2\pi R\,dx)(T - T_\infty) = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t}}$$

(b)

$$\Rightarrow \frac{\partial}{\partial x}\left(k\pi R^2 \frac{\partial T}{\partial x}\right) dx + \dot{q}(\pi R^2)\,dx - h\,(2\pi R\,dx)(T - T_\infty) = \rho c_p (\pi R^2 dx) \frac{\partial T}{\partial t}$$

$\div \pi R^2 dx$

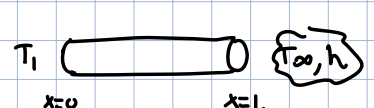
$$\Rightarrow \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \dot{q} + h\left(\frac{2}{R}\right)(T - T_\infty) = \rho c_p \frac{\partial T}{\partial t}$$

$\div \rho c_p$

$$\Rightarrow \frac{\partial T}{\partial t} = \underbrace{\frac{k}{\rho c_p}}_{\alpha,\ \text{thermal diffusivity}} \frac{\partial^2 T}{\partial x^2} + \frac{1}{\rho c_p}\dot{q} - \frac{h}{\rho c_p}\frac{2}{R}(T - T_\infty)$$

$$\Rightarrow \boxed{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}}{\rho c_p} - \frac{2h}{\rho c_p R}(T - T_\infty)}$$

(c)



Boundary conditions

① constant surface temperature: $\boxed{T(0,t) = T_i}$

② convection surface condition: $\boxed{-k \left.\frac{\partial T}{\partial x}\right|_{x=L} = h\,(T(L,t) - T_\infty)}$

Initial condition

$\boxed{T(x,0) = T_i}$

Problem 2

Known:

- $L_A = 30cm$
- uniform volumetric heat generation $\dot{q}$
- left side of wall A is insulated
- $L_B = 10cm$
- $T_\infty = 20°C$
- $T_w = 115°C$
- $T_i = 100°C$

Properties:

- K_A
- $K_B = 30 \frac{W}{mK}$
- $h = 500 \frac{W}{m^2K}$

Assumptions:

- no heat generation within wall B
- no thermal resistance at boundary

Find:

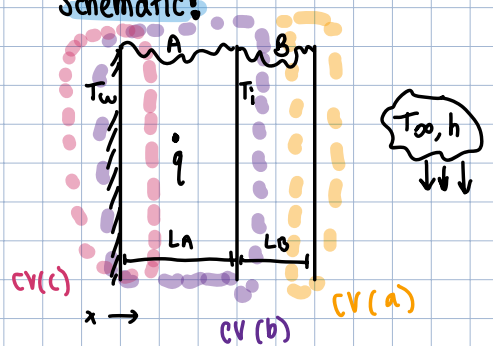
(a) right surface temperature T_s of wall B

(b) uniform volumetric heat generation $\dot{q}$ at SS

(c) K_A

(d) plot temperature profile across wall A & B

Schematic:



Analysis

(a) Control volume around slab b: in orange

$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = E_{st}$$

$$q_{cond} - q_{conv} = 0$$

$$-K\cancel{A}\frac{dT}{dx} - h\cancel{A}(T_s - T_\infty) = 0$$

$$-K\left(\frac{T_s - T_i}{L_B}\right) - h(T_s - T_\infty) = 0$$

$$-\frac{KT_s}{L_B} + \frac{KT_i}{L_B} - hT_s + hT_\infty = 0$$

$$T_s\left(-\frac{k}{L_B}-h\right) = -\frac{kT_i}{L_B} - hT_\infty$$

$$\Rightarrow T_s = \frac{\left(-\frac{kT_i}{L_B} - hT_\infty\right)}{-\frac{k}{L_B}-h} = \frac{-\frac{(30)(100)}{0.1} - 500(20)}{\frac{-30}{0.1} - 500} = 50^\circ C$$

$$\Rightarrow \boxed{T_s = 50^\circ C}$$

(b) at SS: $\dot{E}_{st} = 0$ ← pink CV

$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = 0$$

$$-k\frac{dT}{dx} + \dot{q}L_A = 0$$

$$-k_B\frac{(T_s - T_i)}{L_B} + \dot{q}L_A = 0$$

$$\Rightarrow \dot{q} = \frac{-k_B}{L_A L_{AB}}(T_s - T_i) = \frac{-30}{(0.3)(0.1)}(50-100) = \boxed{50{,}000\ \frac{W}{m^3 K}}$$

(c) heat equation:

$$k_A\frac{\partial^2 T}{\partial x^2} = -\dot{q}$$

$$\int \frac{\partial^2 T}{\partial x^2} = \int -\frac{\dot{q}}{k_A}$$

$$\int \frac{\partial T}{\partial x} = \int -\frac{\dot{q}}{k_A}x + C_1$$

$$T = \frac{-\dot{q}}{2k_A}x^2 + C_1 x + C_2$$

→ use BCs

$$\frac{\partial T}{\partial x} = -\frac{\dot{q}}{k_A}x + C_1 = 0\Big|_{x=0} \rightarrow C_1 = 0$$

$$T(x = L_A = 0.3) = \frac{-50{,}000}{2k_A}(0)^2 + C_2 = 115 \rightarrow C_2 = 115$$

Boundary conditions

$$\frac{\partial T}{\partial x}\Big|_{x=0} = 0$$

insulated

$$T(x=0) = 115^\circ C$$

→ plug in

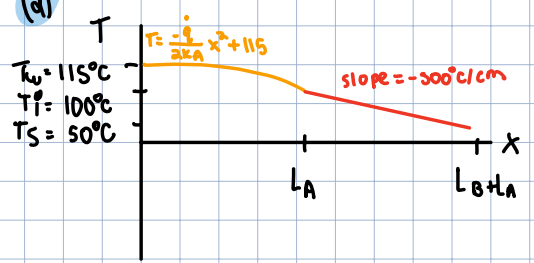
$$T = \frac{-\dot{q}}{2k_A}x^2 + 115 = \frac{-50{,}000}{2k_A}x^2 + 115$$

→ use $T(x = L_A = 0.3) = 100°C$

$$100 = -\frac{50{,}000}{2k_A}(0.3)^2 + 115$$

$$\Rightarrow \boxed{k_A = 150\ \frac{W}{mK}}$$

(d)



finding slope for linear part:

$$\frac{T_s - T_i}{L_B} = \frac{50 - 100}{0.1} = -500°C/cm$$

slope for linear part

Problem 3

Known:

- $k = 25\ \frac{W}{mK}$
- $k_{TBC} = 1\ W/mK$
- $R'' = 10^{-4}\ m^2K/W$
- $T_{\infty,o} = 1616K$
- $T_{\infty,i} = 400K$
- $h_o = 1000\ W/m^2K$ [hot gas]
- $h_o = 500\ W/m^2K$ [cooling air]
- $L_b = 5mm$
- $T_{max} = 1200K$

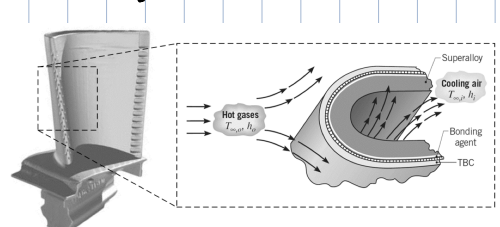
Properties:

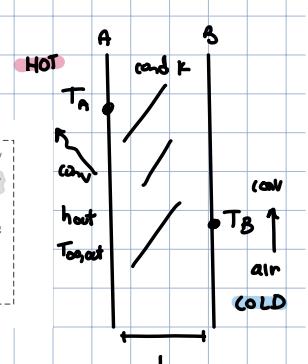
$L_o = 5mm$

Assumptions:

- turbine blade @ SS
- neglect thermal radiation

Schematic:





Find:

(a) exterior surface temperature of super alloy blade, draw thermal resistence network

(b) exterior surface temp of super alloy blade, draw thermal resistence network for TBC applied ($L_{TBC} = 0.5mm$ thick)

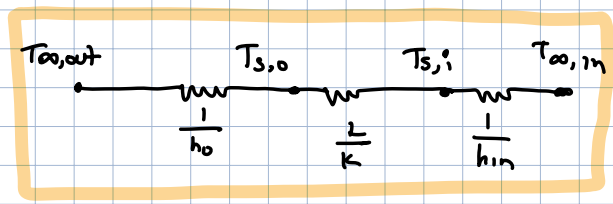
Analysis:

(a) TBC not applied

$$R''_{conv,o} = \frac{1}{h_o} = \frac{1}{1000} = 0.001 \frac{m^2K}{W}$$

$$R''_{cond,o} = \frac{L_b}{k_b} = \frac{0.005}{25} = 0.0002 \frac{m^2K}{W}$$

$$R''_{conv,i} = \frac{1}{h_i} = \frac{1}{500} = 0.002 \frac{m^2K}{W}$$



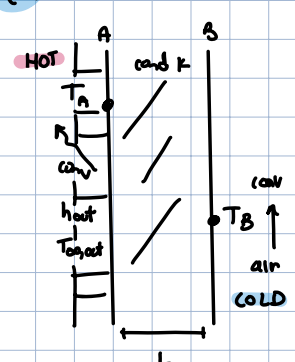
$$\Rightarrow R_{tot} = \frac{1}{h_o} + \frac{L}{k} + \frac{1}{h_{in}} = 0.001 + 0.0002 + 0.002 = 0.0032 \frac{m^2K}{W}$$

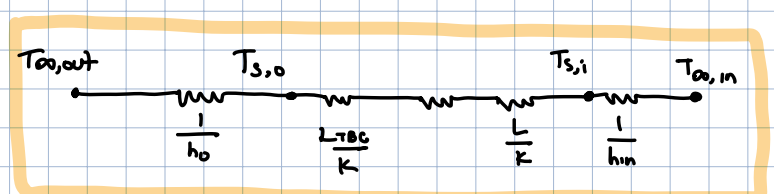
$$\frac{T_{\infty,out} - T_{\infty,in}}{R_{tot}} = \dot{Q} \longrightarrow \dot{Q} = 380000 \frac{W}{m^2}$$

$$T_{s,o} = T_{\infty,o} - \dot{Q}R''_{conv,o} = 1616 - 380000(0.01) = \boxed{1236K}$$

$T_{s,o}$ (1236 K) > T_{max}(1200K) ⇒ blade cannot be maintained below that limit

(b)





$$R''_t = 0.0001 \frac{m^2K}{W}$$

$$R''_{cond,TBC} = \frac{0.0005}{1} = 0.0005 \frac{m^2K}{W}$$

$$\Rightarrow R''_{tot} = 0.001 + 0.0005 + 0.0001 + 0.0002 + 0.002 = 0.0035 \frac{m^2K}{W}$$

$$\Rightarrow \dot{Q} = \frac{1616 - 400}{0.0038} = 320000 \frac{W}{m^2}$$

$$\Rightarrow T_{b,i} = 1616 - 320000(0.001 + 0.005 + 0.001) = \boxed{1104K}$$

⇒ T (1104K) < T_{max}(1200K) ⇒ superalloy blade CAN be maintained below the limit